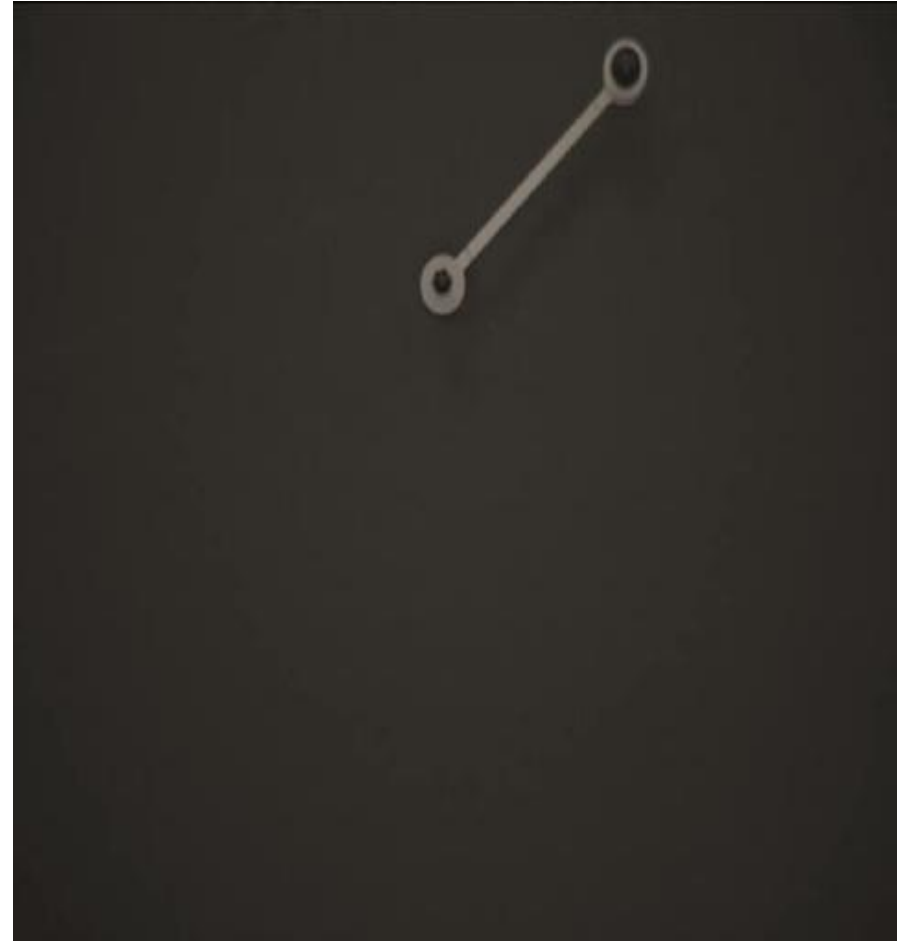


Lab 2: measuring a single pendulum

From high-speed video to a defensible model comparison

Physical model, finite-angle behavior, camera measurement, tracking, and Pre-Lab 2.



What you will do in Lab 2

Release pendulum from rest at three angle ranges.

Record motion with high-speed camera.

Track bob and convert $x(t)$, $y(t)$ into $\theta(t)$.

Compare period and frequency with model.



Physical system and coordinate

M: bob and attached hardware mass

m: uniform bar mass

l: pivot-to-bob-center distance

$\theta(t)$: angle from downward vertical

$\omega(t) = d\theta/dt$; every release begins with $\omega(0) = 0$.

Ideal point-mass pendulum

$$M\ell^2 \, d^2\theta/dt^2 = -Mg\ell \sin \theta$$

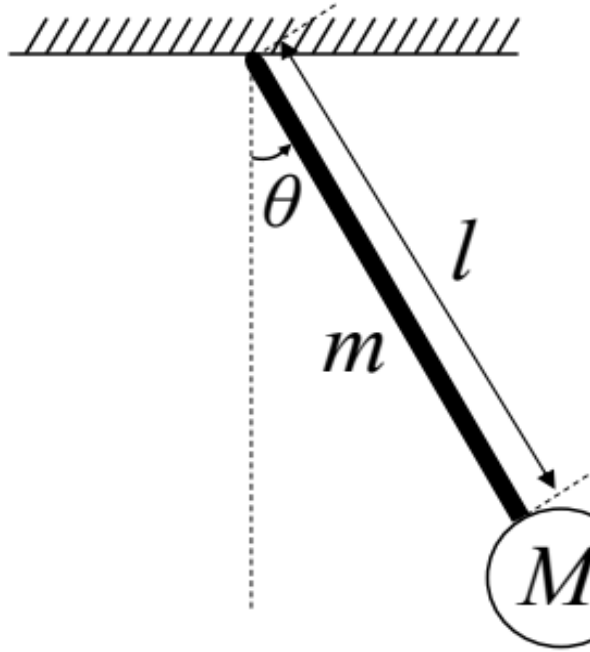
$$d^2\theta/dt^2 + (g/\ell) \sin \theta = 0$$

Gravity drives the bob toward $\theta = 0$.

The sine term makes exact motion nonlinear.

M cancels only because this ideal model treats the bar as massless.

The bar mass changes the model



Moment of inertia about the pivot

$$I_O = M\ell^2 + \frac{1}{3}m\ell^2$$

Gravity torque

$$\tau_g = -(Mg\ell + \frac{1}{2}mg\ell) \sin \theta$$

The bob acts at l . The uniform bar acts through its center of mass at $l/2$.

Compound-pendulum equation

$$(M\ell^2 + \frac{1}{3}m\ell^2) d^2\theta/dt^2 + (Mg\ell + \frac{1}{2}mg\ell) \sin \theta = 0$$

Small-angle approximation

$$\omega_n^2 = g(M + m/2) / [\ell(M + m/3)]$$

$$f_n = \omega_n/(2\pi) \quad T_0 = 2\pi/\omega_n$$

Measure M , m , and ℓ for the apparatus.

The point-mass formula applies only when bar mass is negligible.

Mass distribution changes inertia and gravity torque by different amounts.

Why large initial angle changes period

At large angle, $\sin(\theta)$ differs from θ .

Exact restoring torque is weaker than the linear model predicts at the same angle.

Finite-angle motion takes longer to complete a cycle.

Use the nonlinear equation from Pre-Lab 2 for all three releases.



Energy during a swing

$$E = \frac{1}{2}(M\ell^2 + \frac{1}{3}m\ell^2)(d\theta/dt)^2 + (Mg\ell + \frac{1}{2}mg\ell)(1 - \cos \theta)$$

At release theta is largest and omega = 0.

Near the bottom theta = 0 and speed is largest.

Air drag and pivot friction cause measured amplitude and energy to decrease.

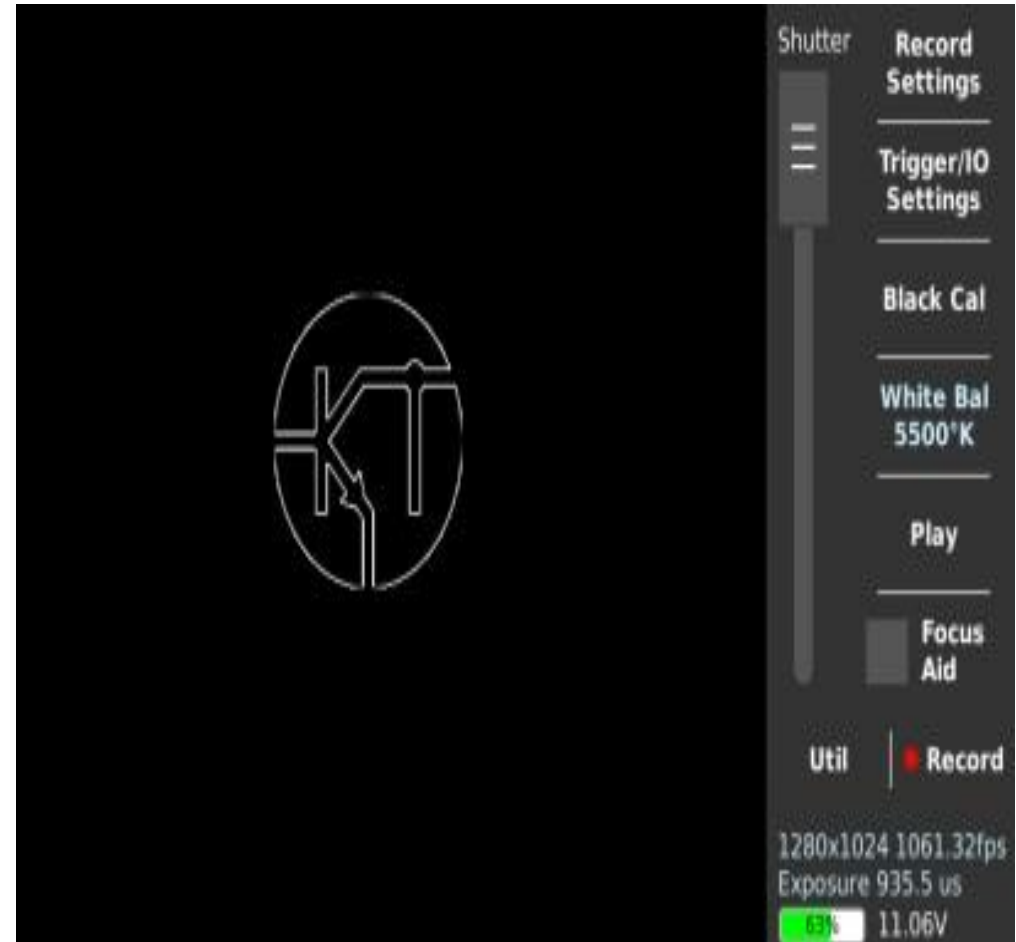
What the camera actually measures

Camera stores frames, not angle directly.

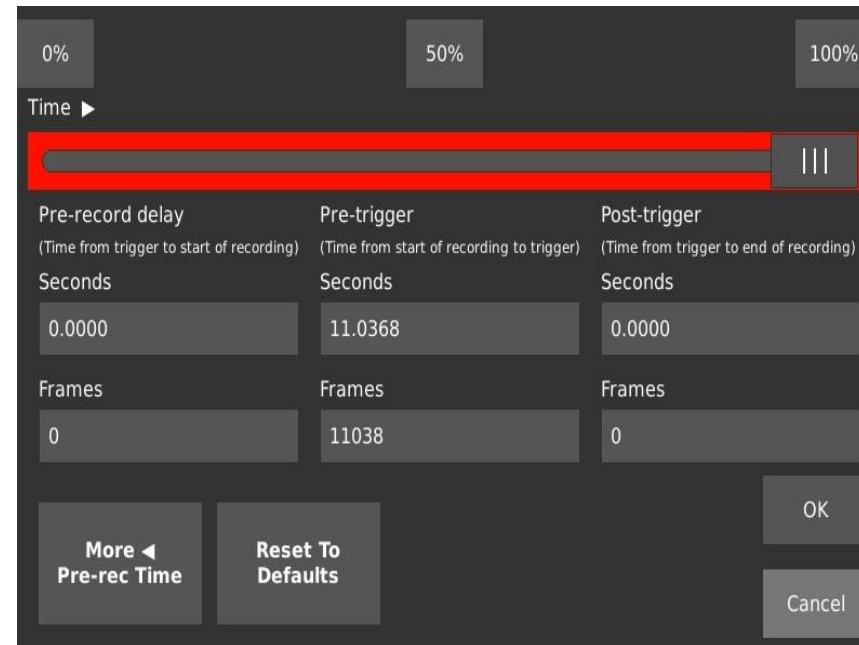
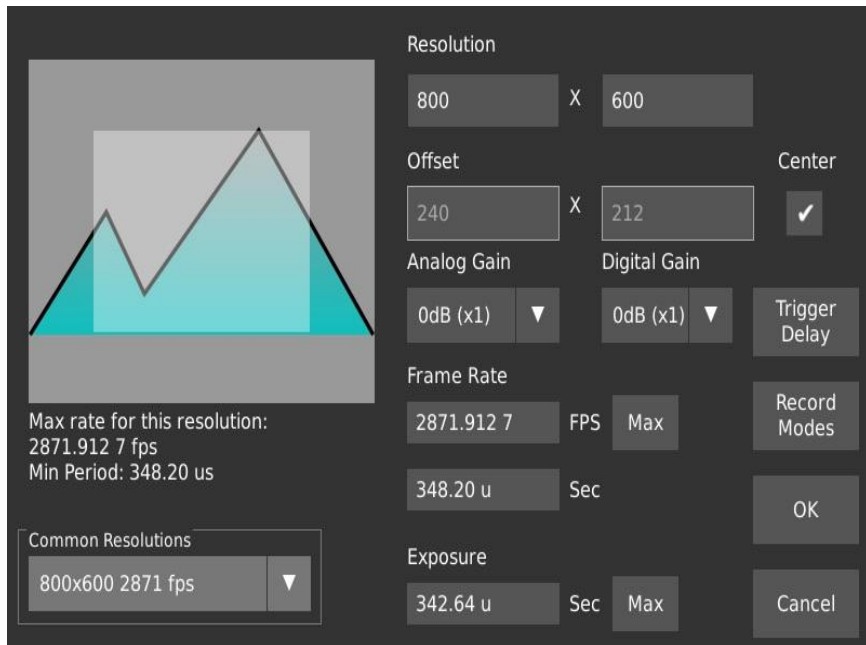
Every frame contains pivot and bob positions.

Useful record needs full motion in frame, sharp contrast, and visible bob center.

Frame rate becomes time spacing of measurement.



Camera settings and useful video



Record actual frame rate.

Use trigger delay that includes release and cycles.

Test framing before experiment.

Smooth video does not prove accurate tracking.

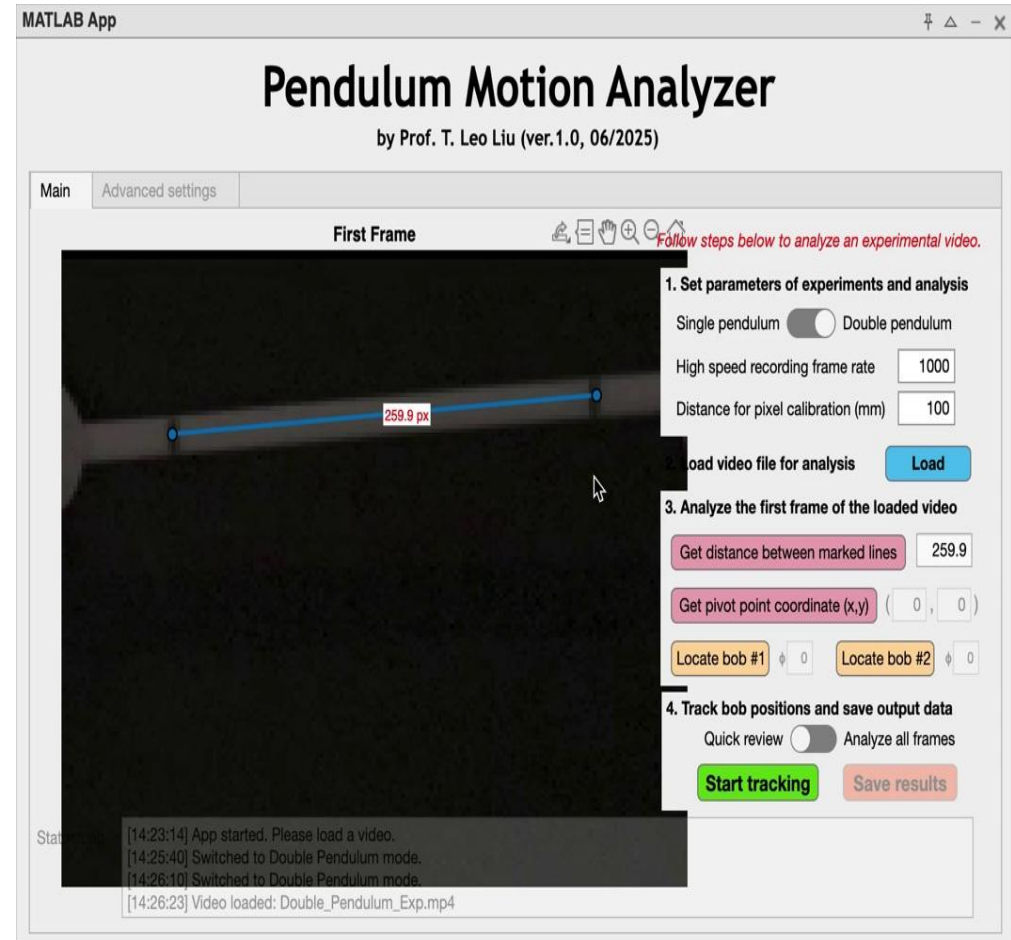
From pixels to coordinates

Calibration marks have known separation, normally 100 mm.

Calibration gives pixel-to-millimeter scale.

Do not move camera or change zoom after calibration.

App saves time and bob coordinates.



From coordinates to a signed angle

$$\theta(t) = \text{atan2}(x_{\beta} - x_p, y_{\beta} - y_p)$$

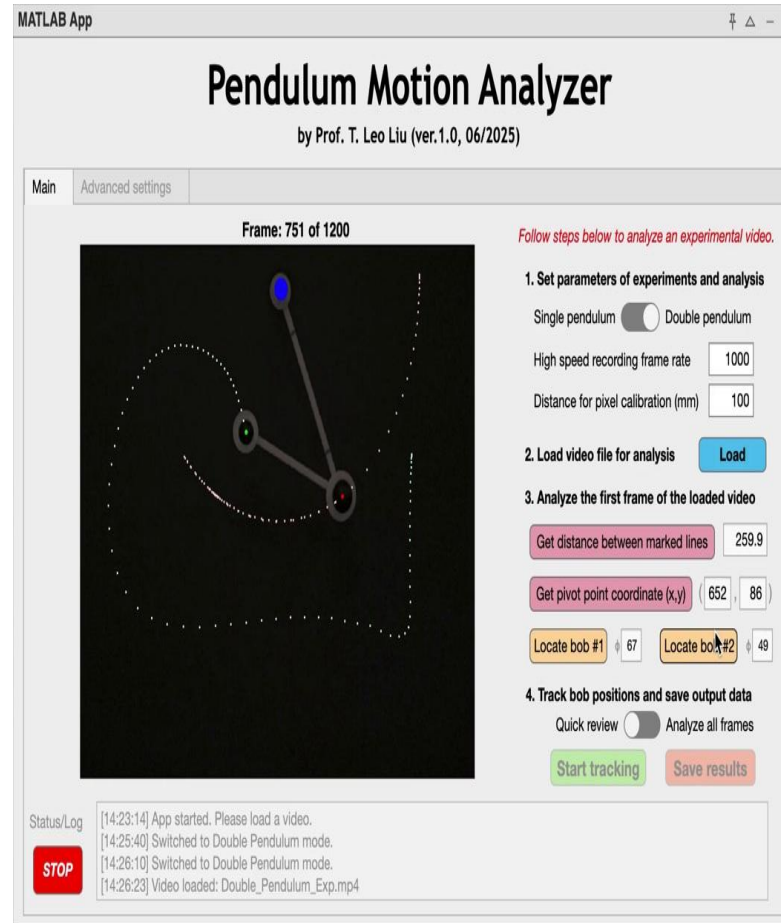
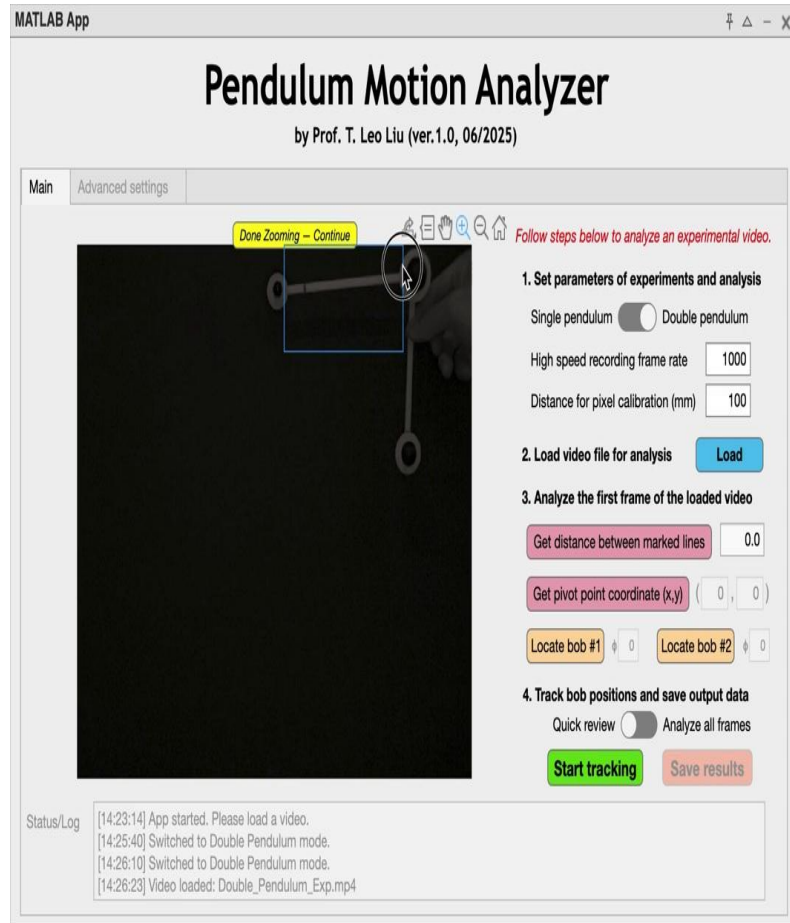
For image coordinates, x points right and y points down.

The formula gives $\theta = 0$ when the bob is directly below the pivot.

Verify zero and sign using one known frame before batch analysis.

Use radians for modeling and derivatives.

Tracking workflow in MATLAB



Select Single pendulum.

Enter actual camera frame rate.

Calibrate, select pivot and bob, then Quick Review.

Run Analyze All Frames only after marker follows bob.

Why frame rate belongs in report

$$f_n = f_s/2 = 500 \text{ Hz} \quad (f_s = 1000 \text{ frames/s})$$

For $l = 0.82 \text{ m}$, one cycle contains roughly 1800 frames.

The rate greatly exceeds the Nyquist minimum for the pendulum.

Wrong frame rate gives wrong time, frequency, and angular velocity.

Angular velocity and noise

$$\omega(t) = d\theta(t)/dt$$

Differentiation magnifies frame-to-frame error.

Inspect $\theta(t)$ before interpreting $\omega(t)$.

State smoothing method and show it does not hide tracking problems.

Maximum speed occurs near the bottom.

Period from time history and spectrum

Time domain: measure several cycles between same-direction peaks.

$$T = (t_{k+n} - t_k)/N \quad f = 1/T$$

Frequency domain: remove mean, compute FFT, identify dominant peak.

Report record duration T_r and frequency-bin spacing.

$$\Delta f = f_s/N_s = 1/T_r$$

Time-domain period and spectral frequency should agree within resolution.

Lab 2 trial matrix

Trial A: θ_0 below 10 degrees. Closest test of small-angle theory.

Trial B: θ_0 from 10 to 90 degrees. Observe period change.

Trial C: θ_0 from 90 to 180 degrees. Expect strong nonlinearity; keep bob in frame.

All releases start from rest. Record θ_0 from protractor.

Data quality before accepting video

Bob remains in frame.

Bob edge and center are visible without glare.

Pivot and calibration marks are visible.

Release begins from rest.

Quick Review follows bob for nearly all frames.

Notes include file name, angle, frame rate, and calibration distance.

Pre-Lab 2: what you submit

One Word or PDF before laboratory section.

Task 1: compare point-mass and compound-pendulum frequencies.

Task 2: predict behavior before running the supplied simulation.

Task 3: estimate periods and explain finite-angle behavior.

Tasks 4-6: explain coordinates, frame rate, and the laboratory plan.

Evidence for combined Lab 2 and 3 report

Schematic: pivot, l , M , m , θ , and positive direction.

Table: measured parameters and analytical, nonlinear, time-domain, and FFT frequencies.

Plots: $\theta(t)$, $\omega(t)$, and spectra with units.

Discussion: separate model limitations from measurement errors.

In-class prediction

For the same apparatus, which release has the longest period?

- A. 8 degrees
- B. 35 degrees
- C. 105 degrees

Explain using the exact sine restoring torque.

Before leaving laboratory

Save and back up every video and MAT output.

Verify tracking for every trial.

Record initial angle, frame rate, calibration distance, l , M , m , and file names.

Check one $\theta(t)$ plot and one spectrum with the GTA.

Key ideas for Lab 2

The real apparatus is a compound pendulum because the bar has mass.

Initial angle changes the nonlinear period.

Camera coordinates require calibration, a fixed pivot, and a verified sign convention.

Compare analytical theory, nonlinear simulation, time history, and spectrum.